Sai Sridhar Tarra
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Education

- **University of Edinburgh** — Edinburgh, UK
  MSc in Statistics with Data Science — *September 2026 – August 2027*
- **Osmania University** — Hyderabad, India
  B.E. in Mechanical Engineering; CGPA: 8.74 — *June 2020 – June 2023*

Experience

- **Accenture** — Bengaluru, India
  Software Engineer — *Nov 2023 – July 2026*
  - **C++ Backend Optimisation**: Profiled and optimised performance-critical C++ backend services for a banking client using STL algorithms and memory-efficient data structures, reducing processing time by 25%.
  - **Python & FastAPI Development**: Built RESTful APIs with Python, FastAPI and PostgreSQL for internal banking services, optimising queries and request handling to cut p95 latency by 20% under concurrent load.
  - **Testing & Reliability**: Wrote unit and integration tests for production banking systems, debugged backend defects with Linux tooling, and contributed to code reviews and Git-based CI/CD workflows.

Projects

- **Huddle: Multiplayer AI Coding Agent Platform** | *TypeScript, Python, PostgreSQL, Docker, WebSockets*
  - Built a multiplayer workspace where a team watches and steers one AI coding agent live, with driver hand-off, queued suggestions and comments.
  - Designed an event-sourced backend: 40+ versioned event types stored append-only in PostgreSQL, reduced into UI state and streamed over a WebSocket server written from scratch to RFC 6455.
  - Implemented the LLM agent loop in Python with tool calling, sandboxing each session in Docker (no network, dropped capabilities, resource limits) with human approval for risky actions; 77 automated tests, self-hosted with HTTPS and nightly backups.
- **MultiLingua** | *PyTorch, XLM-RoBERTa, FastAPI, Docker, MongoDB*
  - Fine-tuned XLM-RoBERTa in PyTorch for cross-lingual sentiment analysis across English, Hindi, and Spanish, reaching 88% accuracy.
  - Added support for code-mixed text (e.g. Hindi–English) so a single model handles mixed-language input without per-language pipelines.
  - Containerised the FastAPI REST service with Docker and logged predictions to MongoDB, sustaining 500+ requests/min.

Technical Skills

Languages: C++, Python, SQL, JavaScript, TypeScript
Backend Engineering: FastAPI, REST APIs, WebSockets, OOP, Data Structures & Algorithms, Asynchronous Programming
Databases: PostgreSQL, MongoDB, SQL Query Optimisation, Database Indexing
Software Engineering: Git, Linux, Unit Testing, Integration Testing, Debugging, Code Reviews
Cloud & DevOps: Docker, CI/CD, AWS
Machine Learning: PyTorch, TensorFlow, Scikit-learn, Hugging Face, NLP, LLM Agents, Statistical Modelling

Achievements & Certifications

- **Coursera ML & Deep Learning Specializations (DeepLearning.AI)**: supervised/unsupervised learning, CNNs, RNNs, LSTMs, Transformers.
- **Competitive Programming**: LeetCode – tarrasridhar1154 – Solved 1000+ problems (DSA, Algorithms, SQL); Codeforces Specialist.